

CHE 317 Introduction to Chemical Engineering Analysis

Spring 2025

Material and Energy Balances form the foundation for nearly all future chemical engineering courses and analysis. The process of account for mass and energy is akin to “chemical accounting” and in this course we will learn how to formulate the balances that govern systems in which mass and energy are exchanged. Included in this analysis are both reactive and nonreactive systems. An overview of physical behavior will be included to provide an introduction to energy balances. Overall, this course will provide an introduction to the principles and calculation techniques used in the field of Chemical Engineering as well as provide an exposure of the various areas and facets of current Chemical Engineering research

Instructor: Thomas Edison

Office: CPE 4.458 (1-4438)

Office hours: M 2:00 – 3:00 pm (at my office CPE 4.458)

Tu, Th, (5 to 6:30 pm) (Zoom office hour)

Email: tedison@che.utexas.edu

Undergraduate TA:

Undergraduate Tutor:

Lectures: MWF 8:00 to 8:50 am at CPE 2.216

Please be punctual. Pre-Lecture problems will be collected by 8:10 am.

Bring your Calculator to class. Recommended Calculator (TI-Nspire CX-II CAS).

You will be expected to solve (with calculator) system of linear and non-linear algebraic equations. Should be able to do numerical integration and be able do linear and multilinear regression in calculator.

Recitation: F 1:00 – 3:00 at CPE 2.212 (There will be a 50 minute quiz from 1:00 to 1:50 pm
1:55 to 2:50 pm problem solving hour (completed work must be uploaded in canvas)

Optional Books: “Elementary principles of Chemical Processes”, (3rd ed., or 4th ed.) R.M.

Felder, and R.W. Rousseau, John Wiley & Sons, 2005.

“Introduction to Material and Energy Balances”, G.V. Reklaitis,

John Wiley & Sons, 1983.

Final Grade: Pre-Lecture problem 8 % (Every lecture period)

Homework 4 % (Weekly)

Quizzes 20 % (Recitation and Take home)

Exams (Three) 54% (Exam-I 18%, Exam-II 18%, Exam-III 18%)

Final Exam 14 %

In the past the class average for this class is between 72 and 76. Class average is given B-. There are two requirements for a passing grade C-: You should have an exam grade of 60 points and have a course average that will be determined from your class average and standard deviation.

Pre-Lecture Problem and Homework:

Pre-Lecture problem is due before each class period (by 8:10 am). Late pre-lecture problem will not be accepted. Pre-Lecture problem must be neatly presented in the standard chemical engineering format. Please use engineering paper.

Weekly homework problem must be submitted on CANVAS by Friday 1:00 PM (only PDF and EXCEL files are accepted)

Pre-Lecture problem and Homework must be handed in individually, but you are strongly encouraged to work in groups to discuss and solve the problems. Homework problems are graded with an emphasis on effort and setting up the problem, and only selected problems within each assignment will be graded.

Quizzes:

There will be a quiz during each recitation class. Each quiz will have one to two problems. The quiz problems will be similar to the homework problems, and the problems discussed in class.

All Quizzes are closed notes. The quizzes will not be equally weighted; they will be weighted according to their difficulty.

Exams:

There will be three exams during the semester. Exams are closed book and closed notes.

Exam – I Feb 7th (Evening 6 to 9 pm CPE 2.216, 2.220)

Exam – II Mar 7th (Evening 7 to 10 pm CPE 2.216, 2.220)

Exam – III April 11th (Evening 6 to 9 pm CPE 2.216, 2.220)

Final Exam May 4th (1:00 pm – 3:00 pm CPE2.220)

(You have one week after the exam is returned to you to request for re-grading. Re-grading requests should be submitted in writing at the end of that lecture period or during office hour)

The University of Austin provides upon request appropriate academic accommodations for

qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-6441 TTY.

Absence: No excuse for missed exams. If you miss an exam due to health or family reason, your final exam score will be used for the missing exam. Final exam will also have problems from missed exam.

Major Topics that will be covered in this course:

- I) **Material Balance in Non-Reacting Systems**
- II) **Material Balance in Reacting Systems**
- III) **PVT behavior of fluids, EOS**
- IV) **Energy Balance in Nonreacting Systems**
- V) **Energy Balance in Reacting Systems**
- VI) **Batch system (Transient Processes)**

Knowledge, Abilities, and Skills Students Should Gain From this Course:

- Familiarity with chemical process equipment
- Exposure to systems analysis and concepts of unit operations
- Ability to convert between different sets of units
- Ability to write and balance chemical reaction equations using atomic balances
- Ability to check degrees of freedom, linear independence of reactions and atomic balances; able to construct independent chemical reactions from a set of chemical species
- Ability to identify the excess and limiting reagents, and calculate the percent conversion and yield of reactions
- Ability to define steady state and distinguish between open and closed systems
- Ability to solve material balances for systems with or without a chemical reaction; ability to solve a number of interconnected units involving recycle, bypass, and purge
- Ability to utilize of Le Chatelier's principles in equilibrium reactor material and energy balance
- Ability to read P-V-T, P-H diagrams for pure fluids, able to calculate vapor and liquid specific volumes and mass, enthalpies, and the quality of two-phase fluids
- Ability to define an ideal gas, and calculate based on the perfect gas equation

- Ability to apply Dalton's and Amagat's Laws for ideal gas mixtures
- Ability to define vapor pressure, saturation, equilibrium, triple point, dew point,

- Ability to apply the phase rule, use binary T-x-y, P-x-y, x-y diagrams to calculate composition of vapor and liquid in flash drums
- Ability to define heat, work, energy, enthalpy, etc.
- Ability to estimate heat capacities and calculate enthalpy changes, with and without phase changes

- Ability to use steam tables,
- Ability to solve energy balances for open and closed systems, with or without chemical reactions

- Ability to solve simple combined material and energy balances (with or without chemical reactions)
- Ability to apply spreadsheet calculations to mass and energy balance problems